Equilibrium Constants

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1. Progress variable of a reaction

The progress variable,ξ, of a reaction indicates how much the reaction has progressed in the forward direction. Consider an example reaction

$$2H_2 + O_2 = 2H_2O$$

If a system consists of N_1, N_2, N_3 and N_4 moles of H_2, O_2, H_2O and N_2 at time t_1, and at time t_2 the reaction progressed by ξ moles, i.e. each mole of progression consumes 2 mole H_2 and 1 mole O_2, and creates 2 mole H_2O. At time t_2 the system will consists of

$$H_2: N_1 - 2\xi$$
$$O_2: N_2 - \xi$$
$$H_2O: N_3 + 2\xi$$
$$N_2:\ N_4$$

Now let's consider a reaction in general form

$$\nu_1'S_1 + \nu_2'S_2 + \cdots + \nu_K'S_K = \nu_1"S_1 + \nu_2"S_2 + \cdots + \nu_K"S_K \tag{1}$$

where the $\nu_i' and\ \nu_i"$ are called the forward and backward stoichiometric coefficients and S_i indicates the *i*th species. Reaction (1) can also be expressed as:

$$\nu_1S_1 + \nu_2S_2 + \cdots + \nu_KS_K = 0$$

where

$$\nu_i = \nu_i" - \nu_i'$$

is called the net stoichiometric coefficient, or simply the **stoichiometric coefficient**.

It is seen that if Reaction (1) progressed by $d\xi$, the change in the number of moles for each species is

$$dN_i = \nu_i d\xi,\ \ i = 1, 2, \dots, K$$

(2)

2. Equilibrium of an individual reaction at constant T, p

At chemical equilibrium for const T & p, the Gibbs free energy of the system is minimized:

$$dG = -SdT + Vdp + \sum_i g_i dN_i$$

Using eq. (2), we can further have

$$dG = \sum_i g_i dN_i = \left(\sum_i g_i \nu_i\right) d\xi$$

since $dT = dp = 0$, i.e.

$$\frac{dG}{d\xi} = \sum_i \nu_i g_i = 0$$

(3)

Eq. (3) is therefore the condition for the equilibrium of a single reaction at const T & p.

As given in the supplementary notes on mixture properties, the free energy of a species can be related to that at **standard state**, that is, at p = 1atm.

$$g_i(p_i, T) = g_i^0(p^0, T) + R_u T ln\left(\frac{p_i}{p^0}\right)$$

where p^0 is the atmospheric pressure, g_i^0 is the free energy (mole based unit) of the ith species at 1atm, therefore it is only a function of T. Eq. (3) then becomes

$$\sum_i \nu_i \left(g_i^0(p_i^0, T) + R_u T ln\left(\frac{p_i}{p^0}\right)\right) = 0$$

$$\frac{\sum_i \nu_i g_i^0}{R_u T} + \sum_i (\nu_i ln\left(\frac{p_i}{p^0}\right)) = 0$$

$$\frac{\sum_i \nu_i g_i^0}{R_u T} + \ln\left(\prod_i \left(\frac{p_i}{p^0}\right)^{\nu_i}\right) = 0$$

$$\prod_i \left(\frac{p_i}{p^0}\right)^{\nu_i} = \exp(-\frac{\sum_i \nu_i g_i^0}{R_u T})$$

Recall that at equilibrium, *Kp* is defined as:

$$\mathrm{K_p} \equiv \prod_i \left(\left(\frac{p_i}{p^0}\right)^{\nu_i}\right) = \prod_i {p_{ia}}^{\nu_i}$$

where p_{ia} is the partial pressure of the ith species in the unit of atm. Therefore we have:

$$Kp(T) = \exp(-\frac{\sum_i \nu_i g_i^0}{R_u T})$$

(4)

Using ideal gas law,

$$p_i = c_i R_u T$$

where c_i is the mole concentration of the ith species, we can readily obtain *Kc* as

$$Kc \equiv \prod_i c_i^{\nu_i} = \frac{Kp(T)}{(R_u T/p^0)^{\sum_i \nu_i}}$$

(5)

Note that the temperature dependent rates of a reversible reaction are related by *Kc*

$$Kc = \frac{k_f}{k_r}$$

where

$$k_f = AT^n \exp(-\frac{E_a}{RT})$$

(6)

for a typical bimolecular reaction. Three parameters A, n, and E_a need to be specified to determine k_f. The reverse rate, k_r, of the reaction is typically evaluated using eq.(6).